



EXPLANATION

$V4 = \{ I, \sigma_v, C2, \sigma_h \}$

"Thinking examines its own frame."

Informative / Explanatory

Owl Semaphore — Explanation

Informative Companion — Origin Story, Archetype Rationale, Accessibility

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OWL SEMAPHORE SYSTEM — MASTER PROOF

**Owl Semaphore v2.0.0-rc — Final Composed Badges (Presentation Layer)**Owl-only V4 tested master + per-state palette meander/ring. Mathematical master remains the transparent owl-only PNG.

Presentation-layer composed badges: per-state palette meander/ring around the V_4 -tested owl-only master. The mathematical master is the owl-only transparent PNG (shown below); the surrounding circle is editorial.

OWL-ONLY MATHEMATICAL MASTER — V_4 -TESTED SOURCE

These are the V_4 -tested owl-only PNGs (owl + human-selected gold branch). They are the algebraic master; tests/test_v2_assets.py gates them. The surrounding circle/meander above is presentation only and is not part of the V_4 transform input.

(Informative companion to the System Specification — v2.0.0-rc)

This document is **informative**, not normative. It tells the origin story, the design rationale, and the audience reasoning that produced the Owl Semaphore. The normative algebra and asset rules live in `OWL-SEMAPHORE-SYSTEM.md` and the four state specifications. Where this document offers warmer wording (especially for METACOGNITIVE), the normative spec retains the scientifically precise version.

1. What the Owl Semaphore is, in one paragraph

The Owl Semaphore is a four-state visual system for marking how a claim, document, dataset, or finding should be evaluated before belief, challenge, or action. The four states — NORMATIVE, NON-NORMATIVE, CRITICAL, METACOGNITIVE — form the Klein four-group V_4 under composition of reflections and 180° rotation. Each state is encoded redundantly through color,



orientation, and a textual label so that the state identity remains recoverable when any one channel fails. In a single human sentence: **four owls tell the reader what kind of thinking they are looking at — standard, exploration, inversion, or self-audit.**

2. Where the Owl Semaphore came from — DNS Tool

The Owl Semaphore was not invented to decorate pages. It grew inside the DNS Tool — a long-running research and operations effort that needed to mark, with discipline, the difference between what is observed and what is inferred, between what is asserted and what is audited, and between what is standard and what is exploratory.

DNS Tool already separated several things its operators had to keep separate:

- **observed fact vs. analytical inference** (DNS Tool confidence framework)
- **assertion vs. audit** of standards and resolver behavior
- **success vs. rate-limited vs. error vs. partial** result states for queries
- **standard vs. operational evidence** in the research corpus (DNS Tool corpus)

DNS Tool’s “good net citizen” posture also imposed minimal-footprint queries, adaptive rate awareness, documented external data sources, and honest rate-limit reporting. Every part of that posture is an epistemic discipline before it is a network discipline. The Owl Semaphore is the visual-algebra layer that makes that discipline legible to a reader who is not inside the codebase.

In the corpus, the Owl Semaphore had already begun to function as an active classification legend with the lines:

- “This is the standard.”
- “This reflects the standard.”
- “This inverts the standard.”
- (a fourth state describing the audit of the frame itself)

That fourth line — the audit of the frame — is what this release tightens. The earlier wording, “*This audits the standard,*” was insufficient because the audit at this state is **not directed at an external standard**. It is directed at the observer’s own evaluative frame. We say so explicitly now: **the observer audits the frame**, or, in plain English, **thinking examines its own frame**.

3. Why four states, not two

Standards culture is comfortable with two states: normative and non-normative. RFC 2119 / RFC 8174 give a precise vocabulary for the normative/non-normative split when the keywords appear in all capitals (RFC 2119; RFC 8174). ISO/IEC Directives Part 2 and W3C Recommendations work in a similar register.

Two states are useful, but they are not enough for the actual congruence problem the DNS Tool faces. A claim, an artifact, or a finding can fail in at least four distinct ways:



1. It can fail to follow the standard it claims to follow. (Need: a baseline reference — **NORMATIVE**.)
2. It can be a legitimate, structured alternative to the standard. (Need: a state for **NON-NORMATIVE** reflection.)
3. It can be deliberately inverted against the baseline for adversarial or falsification analysis. (Need: a state for **CRITICAL** inversion — red-team / falsification.)
4. The observer can be the failure mode — the analyst’s frame of evaluation is what is filtering the truth out of the data. (Need: a state for **METACOGNITIVE** frame-audit — *thinking examines its own frame*.)

The two-state register collapses (3) and (4) into “not normative,” which loses both the adversarial register and the self-audit register. The four-state register keeps them separate so the reader can act differently in each one.

4. Why V_4 — the math

The four states need to *close* under composition: doing any two of them in succession must land you back inside the same set. Otherwise the system would leak into unmarked, unspecified states, and the discipline would erode.

The smallest finite group that fits the four-state behavior we need is the Klein four-group V_4 (Wikipedia; nLab). In the visual plane it is realized by the symmetry group of a non-square rectangle: the identity, vertical reflection, horizontal reflection, and 180° rotation. It is a **finite subgroup of the orthogonal group $O(2)$ isomorphic to V_4** (equivalently, the dihedral group D_2); it is **not** $O(2)$ itself (Knill, Harvard Math 22b, Unit 8: $O(2)$).

What V_4 buys us:

- **closure** — composing any two states yields a state already in the set
- **reversibility** — every state is its own inverse ($g^2 = I$)
- **four stable transforms** — no more, no fewer
- **a Cayley table** that fixes how transitions compose, eliminating arbitrary “fifth states”

What V_4 does **not** buy us: it does not guarantee that the four labels carry the right epistemic meaning, and it does not by itself guarantee correctness or security of any artifact tagged with an owl. Group structure is closure, not truth (nLab). The interpretive rules in the state specs do that work.

5. Why the owl

The Owl of Athena carries an old human association with wisdom, vigilance, and nocturnal pattern recognition. Used loosely, that association would be decorative. In the Owl Semaphore



it is constrained by math: the owl is the carrier on which the four V_4 transforms act, and the transforms are what give the owl its epistemic meaning.

The owl is, in Peirce's terms, mostly **iconic** (it visually resembles the wise-observer archetype) but partly **symbolic** (the assignment of color and orientation to state is by convention learned by the reader) (SEP, Peirce's theory of signs). The notation uses several of Bertin's retinal variables — color hue (selective and associative), orientation (selective), and shape (selective) — paired with a textual label (Moody's dual-coding principle) so that the reader gets multiple, independent cues (Bertin 1967/1983; Moody 2009; Wilke, Fundamentals of Data Visualization).

That is also why the badge is constrained to a finite set of orientations and a fixed shared geometry. Decorative variation would dissolve the algebra. The owl is the carrier; V_4 is the alphabet.

6. METACOGNITIVE — thinking examines its own frame

This is the state whose explanatory wording matters most, because its English label is the easiest to misread.

Metacognition is the cognitive-science term for monitoring and regulation of one's own cognitive processes — *thinking about thinking* (metacognitive reflection review, PMC 11368986). It is not synonymous with “reflection” or “philosophy.” It is the deliberate inspection of the **machinery of evaluation** itself, not of the object under evaluation.

The σ_h operator in V_4 — the horizontal reflection that flips the vertical axis — is the visual analogue of that move. The classic embodied demonstration is the **through-the-legs maneuver**: search a room normally, fail to find the target, bend over, and view the same room upside-down through your legs. The room has not changed. You have not changed as an agent. **The evaluative frame has been inverted.** What was being filtered out by your upright-priors becomes visible. That is the operational meaning of the METACOGNITIVE state.

This release adopts the canonical phrasing:

- **Scientific / normative voice:** “*The observer audits the frame.*” (used in OWL-SEMAPHORE-SYSTEM.md, OWL-4-METACOGNITIVE.md, the four-state ledger on every generated PDF, the README state table)
- **Warmer explanatory voice:** “*Thinking examines its own frame.*” (used in this explanation, in teaching materials, in introductory bridges)

The earlier line “*This audits the standard*” is **deprecated** as of v1.3.0-rc and remains deprecated in v2.0.0-rc. It conflated *frame-audit* with *standard-audit* and read as if the auditor and the audited were both external — the opposite of what the state encodes. Two things are worth keeping clear:

- **Metacognitive prompts surface bias; they do not eliminate it.** The cognitive-science literature is explicit on this point; the Owl Semaphore must not claim more than that (metacognitive reflection review, PMC 11368986; ICD 203 analytic standards, §B ODNI).



- **METACOGNITIVE is observer-level, not object-level.** A CRITICAL owl inverts the *object's* assumptions; a METACOGNITIVE owl inverts the *observer's* frame. They are different operators in V_4 and they encode different work.

7. Accessibility — color is not the only carrier

Color matters in this system: gold for NORMATIVE, teal for NON-NORMATIVE, red for CRITICAL, amethyst for METACOGNITIVE. But color cannot do the work alone.

- About **8% of males and 0.5% of females of Northern-European descent** have red-green color vision deficiency; rates vary by population (PMC global review, 12385717).
- **WCAG 2.2 Success Criterion 1.4.1 (Use of Color)** prohibits color from being the only visual means of conveying information (W3C SC 1.4.1).
- **Section 508 §302.3** requires at least one visual mode of operation that does not require user perception of color (section508.gov).
- **CRITICAL** in particular is intentionally low-contrast: a red owl on a warm-red field. If color were the only carrier, CRITICAL would be unrecoverable for red-axis (protan) CVD users. So it cannot be the only carrier.

The Owl Semaphore therefore requires that **every state identity be recoverable from three independent channels**:

1. **color** — the assigned palette
2. **orientation** — the V_4 transform (upright/inverted $\times$ right/left-facing) applied to the canonical owl
3. **textual label and context** — the literal state token (NORMATIVE, NON-NORMATIVE, CRITICAL, METACOGNITIVE) printed alongside the badge, together with the math/quote tuple

This is the project's “color is not the only carrier” rule. It is design-level conformance with WCAG SC 1.4.1; full empirical Level-AA testing (automated checkers, CVD simulation, user testing — see Wilke, *Fundamentals of Data Visualization*) is scoped to a future release.

8. Who the Owl Semaphore is for — audience archetypes

DNS Tool's hardest design problem was not technical. It was that several expert cultures had to read the same artifacts without alienating each other. The Owl Semaphore is one of the answers: a notation that lets each archetype find what it needs without flattening the others. The archetypes are roughly:

- **Hackers and operators** want proof and reproducibility; they read the integrity manifest and the SHA-3-512 hashes.



- **OSINT and intelligence analysts** want sourcing discipline; they read the citations and confidence language. ICD 203 analytic tradecraft is treated as a reference framework, not as a binding external requirement (ICD 203; ICD 208).
- **DNS engineers** want protocol correctness; they read the standards and the rules of engagement.
- **Standards/RFC readers** want normative discipline; they read the labels and the BCP 14 keyword usage.
- **Data scientists** want explicit visual variables and accessible color; they read Bertin and WCAG.
- **Serious operators** want explicit rules for when to stop, escalate, or audit; they read the interpretation doctrine.
- **Public readers** want a memorable shape and a clear story; they read the four-owl ledger.

The four-state algebra was designed to be the smallest set that lets each of these archetypes act differently without breaking the others' rules.

9. How this connects to the larger Intellectual Resistance framework

The Owl Semaphore sits inside a fourfold operating field that DNS Tool uses to keep work congruent:

- **Verification Principle** — verify the claim
- **Carrier Color Theory** — clean the carrier (don't let the medium leak meaning the message did not authorize)
- **Symbiotic Net Citizen Principle** — protect the system
- **Versioning and Provenance Rule** — preserve the path

The Owl Semaphore marks the **state of thought before action**. It is the visible state layer that helps the system stay congruent across claim, carrier, code, consequence, and correction. Where a claim's verification fails, a CRITICAL owl makes that visible. Where an analyst has just inverted their frame to look again, a METACOGNITIVE owl makes that visible. The semaphore does not assert truth; it asserts **what kind of thinking the reader is looking at**.

10. What the Owl Semaphore is not

- **Not a guarantee of truth.** It encodes evaluative position, not correctness.
- **Not a security proof.** Group structure is closure, not security. seL4 demonstrates that critical software can be subjected to machine-checked formal proof (seL4 SOSP 2009; seL4 whitepaper); the Owl Semaphore aspires to analogous discipline (explicit invariants, bounded state spaces, machine-checkable transitions) without claiming the same proof depth.
- **Not an accessibility certification.** It targets WCAG 2.2 Level A intent for SC 1.4.1; empirical Level-AA conformance testing is future work.



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- **Not a moral judgment.** A CRITICAL owl does not call the underlying claim wrong; it says someone has subjected it to adversarial analysis.
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11. Bridges to DNS Tool and Zenodo

- DNS Tool Owl Semaphore page: <https://dnstool.it-help.tech/owl-semaphore>
 - DNS Tool confidence framework: <https://dnstool.it-help.tech/confidence>
 - DNS Tool corpus (where Owl Semaphore functions as the live legend): <https://dnstool.it-help.tech/corpus>
 - DNS Tool publications: <https://dnstool.it-help.tech/publications>
 - Currently published Zenodo record (v1.2.0): <https://doi.org/10.5281/zenodo.19474599>
 - Concept DOI (all versions): <https://doi.org/10.5281/zenodo.19473697>
 - v1.3.0 version DOI: TBD_BY_ZENODO_ON_RELEASE (not yet minted; this branch is a release candidate)
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12. Closing

The Owl Semaphore tells the reader what kind of thinking they are looking at. Four owls, four V_4 transforms, four colors, four orientations, four labels — and one rule that ties the whole thing together: *the system encodes position, not truth*. When the METACOGNITIVE owl is present, the system has just done something the system most needs to be able to do: **think about its own thinking**.



OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER (v2.0.0-rc)



NORMATIVE

$T = I \quad \det = +1$
 $(x, y) \rightarrow (x, y)$
“This is the standard.”
RFC 2119 MUST / SHALL



NON-NORMATIVE

$T = \sigma_v \quad \det = -1$
 $(x, y) \rightarrow (-x, y)$
“This reflects the standard.”
Informative / Advisory (NOTE)



CRITICAL

$T = C_2 \quad \det = +1$
 $(x, y) \rightarrow (-x, -y)$
“This inverts the standard.”
RFC 2119 MUST NOT / SHALL NOT



METACOGNITIVE

$T = \sigma_h \quad \det = -1$
 $(x, y) \rightarrow (x, -y)$
“The observer audits the frame.”
Epistemic / Framework (META)

Accessibility: color is not the only carrier. State identity is recoverable from color + orientation + textual label (WCAG 2.2 SC 1.4.1).